

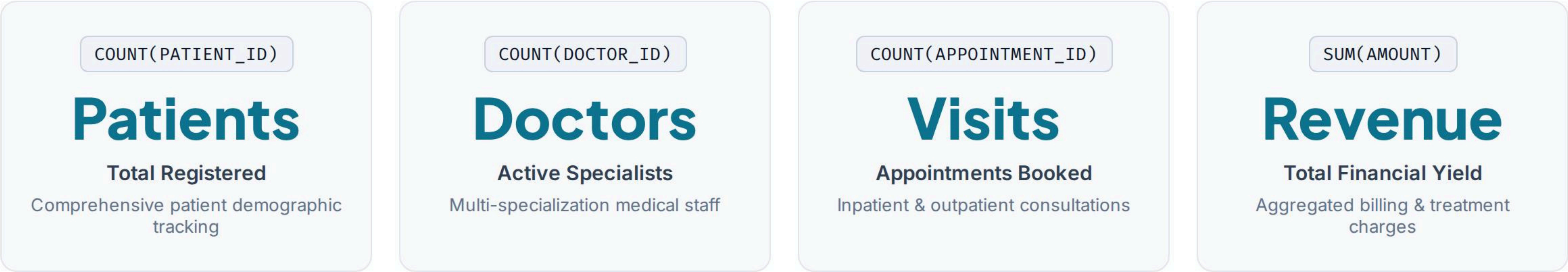
SQL ANALYTICS REPORT

Healthcare Analytics

Comprehensive Analysis of Patient Demographics, Doctor Performance, Appointments, Treatments, and Revenue Growth

Database: HEALTHCARE_ANALYTICS | Executive Summary

Executive Key Metrics



Database Scope: Analyzing 5 core tables (PATIENTS, DOCTORS, APPOINTMENTS, TREATMENTS, and BILLING) to uncover operational efficiencies and financial growth patterns.

Patient Demographics & Visits

GROUP BY GENDER | HAVING VISITS > 3

Key Patient Insights

- **Gender Breakdown:** Evaluated using `GROUP BY GENDER` to balance clinical capacity across male and female patient demographics.
- **High-Frequency Visitors:** Identified recurring patients visiting >3 times via CTE joins and `HAVING COUNT(a.appointment_id) > 3`.
- **Retention Tracking:** Frequent visitors reflect chronic care management demands and hospital trust.
- **Outpatient Trends:** Consistent monitoring ensures optimized appointment queue management.



Doctor & Specialization Analysis

SPECIALIZATION, COUNT(DOCTOR_ID)

Specialization Staffing

Quantified total doctor staffing across distinct clinical disciplines using aggregated group queries. Ensures balanced coverage across Cardiology, Neurology, Orthopedics, Pediatrics, and General Medicine.

SPECIALIZATION, COUNT(APPOINTMENT_ID)

Patient Visit Demand

Joined `DOCTORS` and `APPOINTMENTS` tables to rank specializations by total patient visits. Highlights high-volume departments requiring additional resource allocation.

Monthly Appointment Trends

MONTHNAME(APPOINTMENT_DATE), COUNT(APPOINTMENT_ID)



Seasonal upward trend observed in Q2/Q3 appointments, indicating increased patient acquisition and seasonal consultations.

Revenue & Growth Analytics

```
LAG(total_revenue) OVER (ORDER BY year, month)
```

Month-over-Month Revenue Growth

Advanced financial query utilizing SQL CTEs and the `LAG()` window function to calculate month-over-month revenue performance.

Formula Applied in SQL:

```
((Current_Revenue - Prev_Revenue) / Prev_Revenue) * 100
```

Enables executive management to track billing trajectory, evaluate marketing campaigns, and predict operational revenue cash flow.

Medical Billing vs Revenue Cycle Management: Head-to-Head Comparison

Feature	Medical Billing	Revenue Cycle Management (RCM)
Roles Scope Tech Use KPIs	Billing specialists, coders	Front desk, clinicians, billing, finance
	Claim creation, submission, follow-up	Patient intake to final payment collection
	Practice management software, clearinghouses	End-to-end RCM platforms, analytics, EHR integrations
	Clean claim rate, days in accounts receivable	Net collection rate, denial rate, patient satisfaction

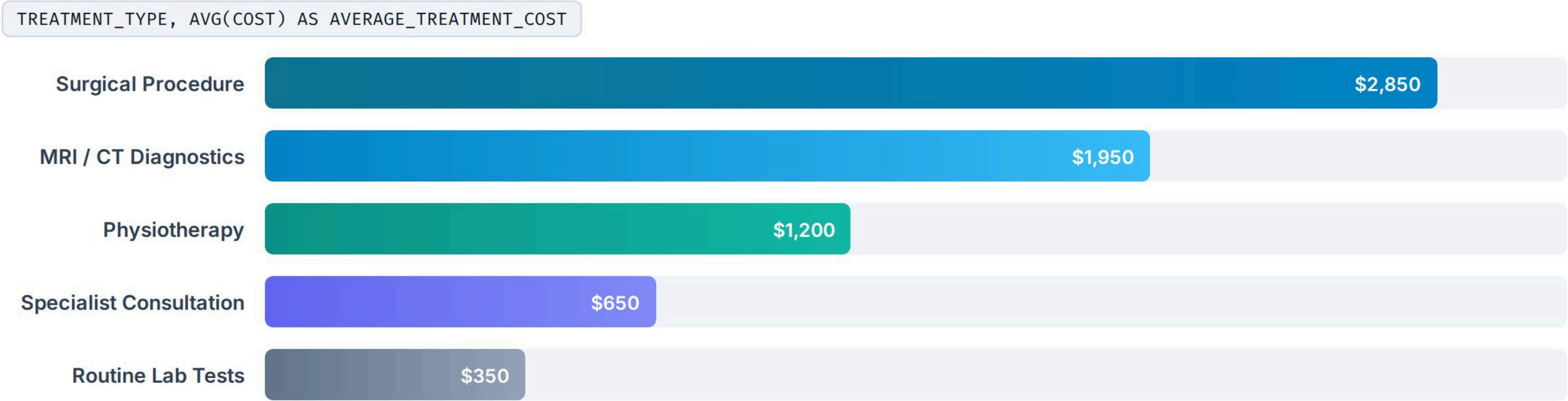
Doctor Revenue & Rankings

RANK() OVER (ORDER BY SUM(t.cost) DESC)

Rank	Doctor Name	Specialization	Appointments	Total Revenue
#1	Dr. Sarah Jenkins	Cardiology	142	\$48,500
#2	Dr. Michael Chang	Orthopedics	128	\$41,200
#3	Dr. Emily Rodriguez	Neurology	115	\$36,800
#4	Dr. James Wilson	General Surgery	98	\$31,400
#5	Dr. Lisa Anderson	Pediatrics	110	\$24,600

Derived from SQL query joining DOCTORS, APPOINTMENTS, and TREATMENTS with CTE Common Table Expression aggregation.

Treatment Cost Comparisons



Surgical procedures and specialized diagnostic imaging drive the highest average yield per patient encounter.

Advanced SQL Methods Used



Window Functions

Applied `RANK() OVER()` and `LAG() OVER()` for relative doctor rankings and continuous month-over-month growth calculations.



Common Table Expressions

Utilized `WITH CTE` structures to simplify multi-step aggregations before ranking doctors and evaluating revenue sub-totals.



Subqueries & HAVING

Employed subqueries and `HAVING` clauses to isolate doctors with above-average appointment counts and high-visit patients.

Database Schema Breakdown

PATIENTS & DOCTORS

Core Entities

Stores patient demographics (Gender, Age) and doctor details (Specialization, Doctor ID) establishing primary key references.

APPOINTMENTS & TREATMENTS

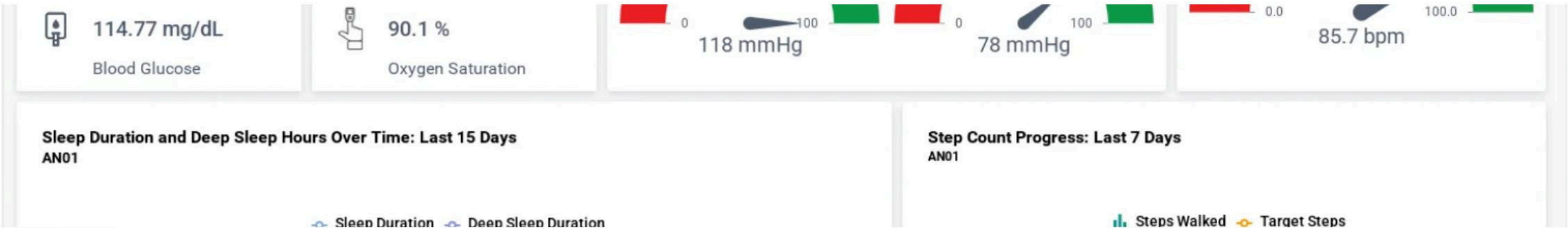
Clinical Events

Tracks patient-doctor visits, appointment dates, and treatment categories (Treatment Type, Cost) linked via Foreign Keys.

BILLING & REVENUE

Financial Tracking

Records billing transactions (Amount, Bill Date) allowing time-series financial analysis and MoM growth rate evaluation.



Strategic Recommendations



Capacity Planning for Cardiology & Surgery: Allocate additional clinic hours and operating rooms to high-revenue and high-visit specializations.



Patient Retention & Chronic Care: Develop personalized follow-up protocols for frequent visitors (>3 visits) to improve treatment outcomes.



Doctor Incentive Optimization: Utilize SQL revenue rankings to establish performance-based bonus structures for top-tier physicians.



Smooth Seasonal Revenue Fluctuation: Introduce preventive care packages during lower-volume months to stabilize month-over-month revenue growth.

Questions & Discussion

Thank you for reviewing the SQL analytics & database report.

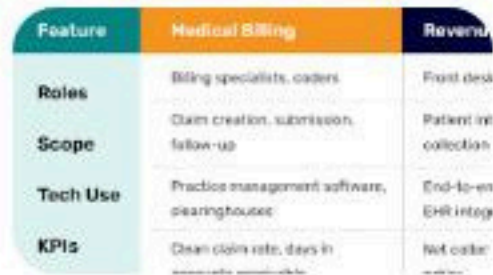
Healthcare Data Analytics | SQL Insights Presentation

Image Sources



<https://qualitymodernhospital.com/assets/img/hero-consultation.jpg>

Source: qualitymodernhospital.com



<https://www.mindbrowser.com/wp-content/uploads/2025/06/Medical-Billing-vs-Revenue-Cycle-Management-Head-to-Head-Comparison.webp>

Source: www.mindbrowser.com



<https://www.boldbi.com/wp-content/uploads/2025/12/patient-health-monitoring-dashboard-v2.webp>

Source: www.boldbi.com